Übungsblatt 5: Integrale

Görkem Akdogan	2120621	Moritz Stiefvater	2124011	Marek Kisek	2020780
Ivo Rickert	2212038	Steven Dörzbach	2211399		
René John	2194387	Simon Schäfer	2123393		

Integral ü5

2170621
Görken
Hudgson

(1)

a) $f(x) = x^n \rightarrow F(x) = \frac{1}{n+1} x^{n+1} + C$

b) $f(x) = 3x^5 - 2x^3 + x^2 + 1$

$\hookrightarrow F(x) = \frac{3}{6}x^6 - \frac{2}{4}x^4 + \frac{1}{3}x^3 + x + C$

d) $f(x) = (2x)^2 = 4x^4 \rightarrow F(x) = \frac{4}{5}x^5 + C$

c ~~d~~) $f(x) = x^{2n} \rightarrow F(x) = \frac{1}{2n+1} x^{2n+1} + C$

e) $f(x) = \ln(x) \rightarrow F(x) = \ln(x) \cdot x - x + C$

f) $f(x) = 1 \rightarrow F(x) = x + C$

g) $f(x) = 0 \rightarrow F(x) = C$

h) $f(x) = \frac{1}{\sqrt{x}} = \frac{1}{x^{\frac{1}{2}}} = x^{-\frac{1}{2}} \rightarrow F(x) = 2 \cdot \sqrt{x} + C$

i) $f(x) = \frac{e}{-x^{-n+1}} = -e \cdot x^{n-1} \rightarrow F(x) = \frac{-e}{n} \cdot x^n + C$

j) $f(x) = e^{ax} \rightarrow F(x) = \frac{1}{a} e^{ax} + C$

k) $f(x) = \cos(x) + \sin(x) \rightarrow F(x) = \cos(x) \cdot t + \sin(x) \cdot t + C$

l) $f(x) = \frac{5}{1+x^2} \rightarrow F(x) = 5 \cdot \arctan(x) + C$

m) $f(x) = \frac{5}{3+3x^2} + \frac{1}{2}x^4 \rightarrow F(x) = \frac{5}{3}\arctan(x) + \frac{x^5}{10} + C$

n) $f(x) = \frac{-2}{\sqrt{1-x^2}} - \frac{1}{\cos^2(x)} \rightarrow F(x) = -2\arcsin(x) - \tan(x) + C$

Integral Ü5

2120621
Görken
Aldogan

① o, r, q + ②

⑥ o) $f(x) = 2\sqrt{x} = 2 \cdot x^{\frac{1}{2}} \rightarrow F(x) = 2 \cdot \frac{2}{3} \cdot x^{\frac{3}{2}} + C = \frac{4}{3}x^{3/2} + C$

r) $f(x) = u'(x) \cdot v(x) + v'(x) \cdot u(x) \rightarrow F(x) = u(x) \cdot v(x) + C$

q) $f(x) = \cos(x) \cdot x + \sin(x) \rightarrow F(x) = \sin(x) \cdot x + C$

② a) $\int_1^2 \frac{1}{x}\,dx \rightarrow \left[\ln(x)\right]_1^2 = \ln(2) - \underbrace{\ln(1)}_{=0} = \ln(2)$ //

b) $\int_1^2 \frac{1}{x^2}\,dx = \int_1^2 x^{-2}\,dx = \left[-x^{-1}\right]_1^2 = -\frac{1}{2} - \left(-\frac{1}{1}\right)$

$= 1 - 1/2 = \underline{\underline{1/2}}$

c) $\int_0^{\pi} \cos(x)\,dx = \left[\sin(x)\right]_0^{\pi} = \sin(\pi) - \sin(0) = 0$ //

d) $\int_{-\pi}^{\pi} \cos(x)\,dx = \left[\sin(x)\right]_{-\pi}^{\pi} = \sin(\pi) - (-\sin(-\pi)) = 0$ //

$\sin(\pi) = 0$, $-\sin(-\pi) = 0$

e) $\int_0^{\pi} \sin(x)\,dx = \left[-\cos(x)\right]_0^{\pi} = \underbrace{-\cos(\pi)}_{=1} - (\underbrace{-\cos(0)}_{=1}) = 2$ //

f) $\int_{-\pi}^{\pi} \sin(x)\,dx = \left[-\cos(x)\right]_{-\pi}^{\pi} = -\cos(\pi) - (-\cos(-\pi)) = 1 - (+1) = 0$ //

g) $\int_0^4 |2x-3|\,dx \neq \left[x^2 - 3x\right]_0^4$ ✗

$= \int_0^{3/2} (3-2x)\,dx + \int_{3/2}^4 (2x-3)\,dx = \left[3x - x^2\right]_0^{3/2} + \left[x^2 - 3x\right]_{3/2}^4$

$= 3 \cdot \frac{3}{2} - \frac{9}{4} + 16 - 12 - \frac{9}{4} + \frac{9}{2}$

$= 4 - \frac{9}{2} + 9 = 13 - \frac{9}{2} = \frac{26-9}{2} = \underline{\underline{\frac{17}{2}}}$

R

L

(4)

$$A(t) = \int_0^t f(x)\,dx$$

$A(t) = F(t) - F(0)$

$A'(t) = F'(t) - 0$

$A'(t) = f(t)$

$\Rightarrow$

$e^{-t^2} = \frac{1}{e}$

$e^{-t^2} = e^{-1}$

$-t^2 = -1$

$t^2 = 1$

$t = \pm 1$

1 Cr

$e^{-x^2} = e^{-1} \Rightarrow -x^2 = -1$

$\Rightarrow t^2 = 1, \quad \underline{t = \pm 1}$

Aufgabe 5 $\int_{-1}^{+1} (x^2 - a)\,dx = 0$

$= \left[\frac{1}{3}x^3 - ax\right]_{-1}^{1} = \left(\frac{1}{3} - a\right) - \left(-\frac{1}{3} + a\right)$

$\left(\frac{1}{3} - a\right) - \left(-\frac{1}{3} - a\right) = \frac{2}{3} - 2a$

$\frac{2}{3} - 2a = 0$

$\underline{a = \frac{1}{3}}$

Aufgabe 6 $\int_{-1}^{1} \tan(\underbrace{3x^5 - x^3 + 7x}_{\text{Punktsymmetrisch}})\,dx = 0$ (ungerade Symmetrie)

Aufgabe 7 $f(x) = \cosh(x) = \frac{e^x + e^{-x}}{2}$

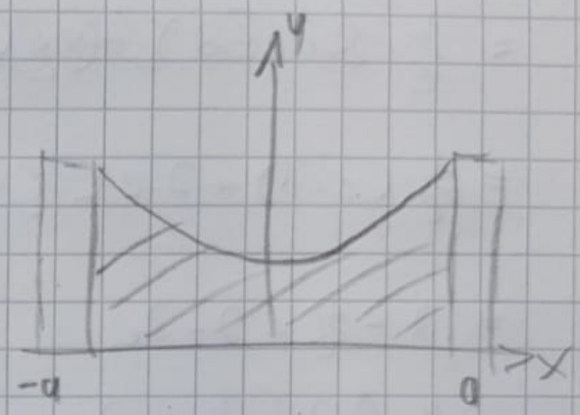
da alle x ungerade

Aufgabe 7 $f(x) = \cosh(x) = \frac{e^x + e^{-x}}{2}$

$$\bar{y} = \frac{1}{b-a} \cdot \int_a^b f(x)\,dx$$

$$\bar{y} = \frac{1}{a-(-a)} \int_{-a}^{a} \cosh(x)\,dx$$

$$= \frac{1}{2a} \int_{-a}^{a} \left(\frac{e^x}{2} + \frac{e^{-x}}{2} \right) dx$$



7

$$= \frac{1}{4a}\left(\int_{-a}^{a} e^{x}\,dx + \int_{-a}^{a} e^{-x}\,dx\right)$$

$$= \frac{1}{4a}\left(\left[e^{x}\right]_{-a}^{a} + \left[-e^{-x}\right]_{-a}^{a}\right.$$

$$= \frac{1}{4a}\left(e^{a} - e^{-a} + \left(-e^{-a} + e^{a}\right)\right)$$

$$= \frac{1}{4a}\left(e^{a} - e^{-a} + e^{a} - e^{-a}\right)$$

$$= \frac{1}{2a}\left(e^{a} - e^{-a}\right)$$

$$= \frac{e^{a} - e^{-a}}{2a} = \frac{\sinh(a)}{a}$$

Aufgabe 8

$f(x) = e^{\frac{1}{2}x-1}$; $g(x) = e^{1-\frac{1}{2}x}$

Schnittpunkt

$e^{\frac{1}{2}x-1} = e^{1-\frac{1}{2}x} \Rightarrow \frac{1}{2}x - 1 = 1 - \frac{1}{2}x \Rightarrow x - 2 = 2 - x \overset{|+2+x}{\Rightarrow} 2x = 4 \overset{:2}{\Rightarrow} x = 2$ //

Fläche zwischen den Fkt. mit Grenzen $\{0, 2\}$

$$\left| \int_{G_1}^{G_2} f(x) - g(x) \right|$$

$$= \left| \int_0^2 e^{\frac{1}{2}x-1} dx - \left(e^{1-\frac{1}{2}x} dx\right) \right|$$

$$\Rightarrow \left| \int e^{\frac{1}{2}x-1} dx - \int e^{1-\frac{1}{2}x} dx \right|$$

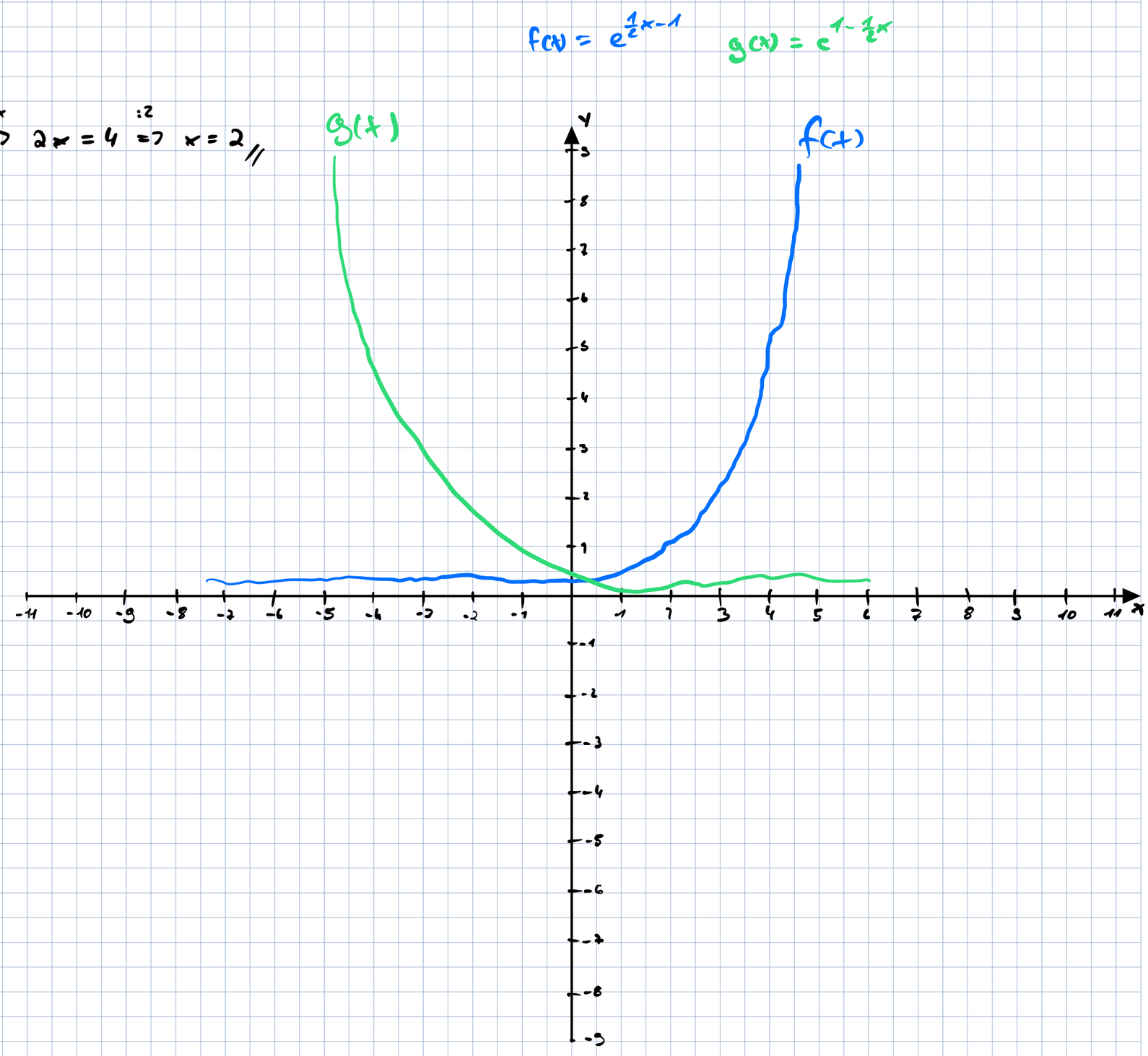
$$\Rightarrow \left[\left| 2e^{\frac{1}{2}x-1} + 2e^{1-\frac{1}{2}x} \right| \right]_0^2$$

$$\Rightarrow \left| 2e^{\frac{1}{2}\cdot 2-1} + 2e^{1-\frac{1}{2}\cdot 2} - \left(2e^{\frac{1}{2}\cdot 0-1} + 2\cdot e^{1-\frac{1}{2}\cdot 0}\right) \right|$$

$$= \left| 2\cdot 1 + 2\cdot 1 - \left(2\cdot\frac{1}{e} + 2\cdot e^{1}\right) \right|$$

$$= 4 - \frac{2}{e} - 2e \quad //$$

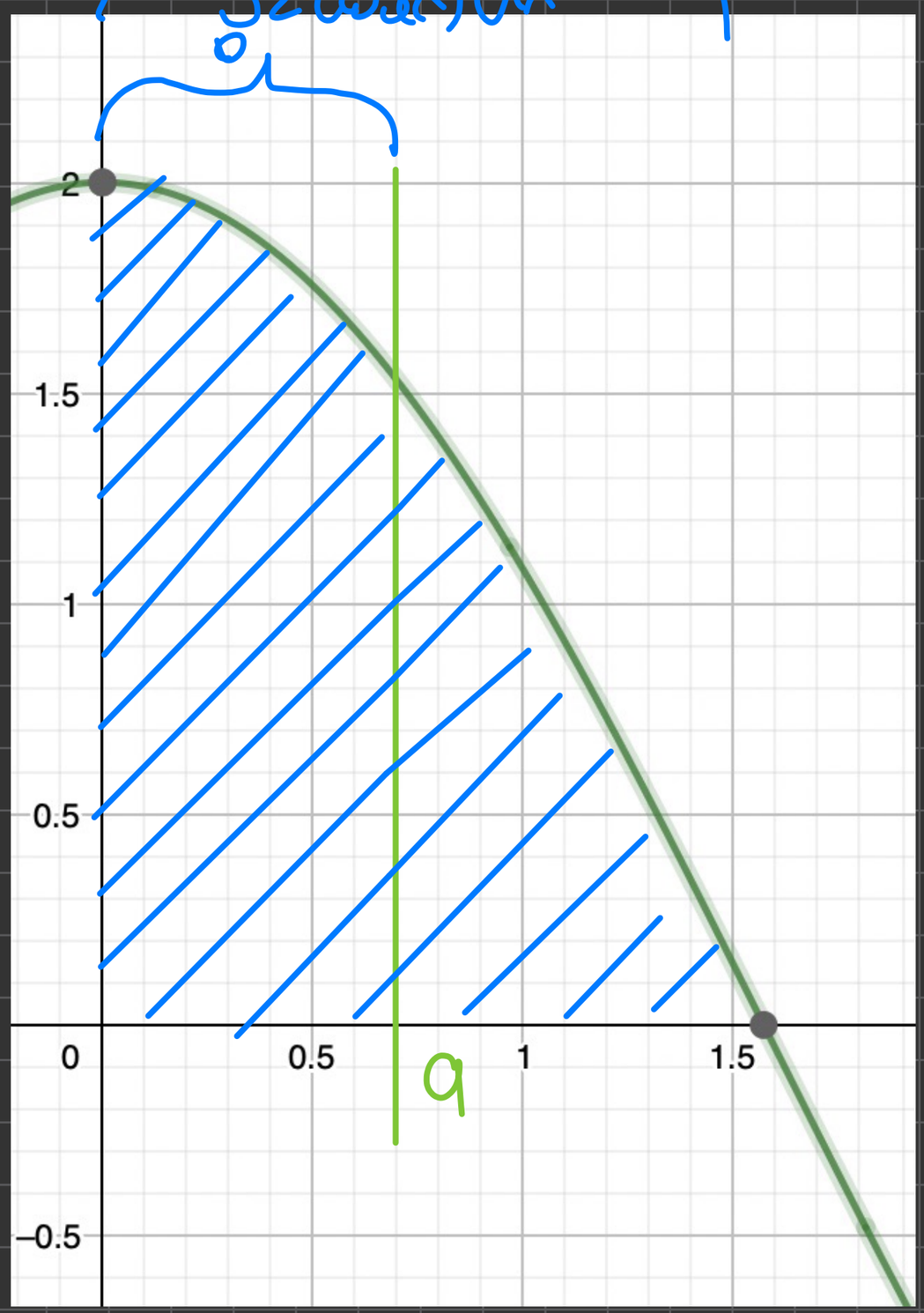
$$\approx 2{,}172 \text{ FE}$$



Aufgabe 9:

$$\int_0^{\pi/2} 2 \cdot \cos(x)\,dx$$

$$\int_0^{q} 2 \cdot \cos(x)\,dx$$



$$2 \cdot \int_0^{a} 2 \cdot \cos(x)\,dx = \int_0^{\pi/2} 2 \cdot \cos(x)\,dx$$

$$\Leftrightarrow \quad 4 \cdot \left[\sin(x)\right]_0^{a} = 2 \cdot \left[\sin(x)\right]_0^{\pi/2}$$

$$\Leftrightarrow \quad 4 \cdot \sin(a) = 2 \cdot \sin\left(\frac{\pi}{2}\right) \qquad /:4$$

$$\Leftrightarrow \quad \sin(a) = \frac{\sin\left(\frac{\pi}{2}\right)}{2}$$

$$\Leftrightarrow \quad \sin(a) = 1/2 \cdot 1 \qquad /\cdot \sin^{-1}$$

~~$a = 1.570796633$~~

$$\Leftrightarrow \quad \sin(a) = 1/2 \quad \Leftrightarrow \quad a = \arcsin(1/2) = \frac{\pi}{6}$$

Aufgabe 10:

$$\int_0^b \sqrt{x}\,dx = 1$$

unbest. Integral

$$\int \sqrt{x}\,dx = \int x^{\frac{1}{2}}\,dx = \frac{2}{3}x^{\frac{3}{2}} + C$$

$$\Rightarrow \left[\frac{2}{3}x^{\frac{3}{2}}\right]_0^b = 1$$

$$\Leftrightarrow \frac{2}{3}b^{\frac{3}{2}} = 1 \qquad / \cdot \frac{3}{2}$$

$$b^{\frac{3}{2}} = \frac{3}{2} \qquad / \sqrt[\frac{3}{2}]{\;}$$

$$b \approx 1{,}31$$

$$= -\frac{1}{2} g t^2 + v_0 t + s_0$$

$$= \int (-g t + v_0) dt$$

$$s(t) = \int v(t) dt$$

$$= -g \cdot t + v_0$$

$$v(t) = \int a(t) dt$$

Aufgabe 44 $s(t) \rightleftarrows v(t) \rightleftarrows a(t)$

Aufgabe 12

$g(x) = mx + b$

$R + m \cdot h = 0 \quad \Big\} \; b = R$

$m = -\frac{R}{h}$

$g(x) = -\frac{R}{h} \cdot x + R$ $\qquad V = \frac{1}{3}\pi R^2 \cdot h$

$b = R$

$V = \pi \cdot \int_0^h \left(-\frac{R}{h}x + R\right)^2 dx$

$V = \pi \cdot \int_0^h \left[\left(-\frac{R}{h}x\right)^2 - 2\left(\frac{R}{h}x \cdot R\right) + R^2\right] dx$

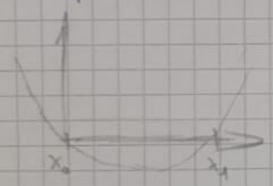
$V = \pi \left(\left(\frac{R}{h}\right)^2 \int_0^h x^2 dx - 2\frac{R^2}{h}\int_0^h x \cdot dx + R^2 \int_0^h dx\right)$

$V = \pi \left(\left(\frac{R}{h}\right)^2 \frac{1}{3}h^3 - 2\frac{R^2}{h} \cdot \frac{1}{2}h^2 + R^2 \cdot h\right)$

$V = \pi \cdot \left(\frac{1}{3}R^2 h - R^2 \cdot h + R^2 \cdot h\right)$

$= \frac{1}{3}R^2 \cdot \pi \cdot h$

Aufgabe 14 $f_t(x) = (1-t)x^2 - tx$



$f_t(x_{0/1}) = 0 \Rightarrow (1-t)x_{0/1}^2 - tx_{0/1} + 0 = 0 \Rightarrow x_{0/1}\left((1-t)x_{0/1} - t\right)$

$x_0 = 0$

$(1-t)x_1 = t \quad | \; x_1 = \frac{t}{1-t}$

$A(t) = -\int_{x_0}^{x_1} f_t(x)\,dx$

$= \int_0^{\frac{t}{1-t}} \left((1-t)x^2 - tx\right) dx$

$= -(1-t)\int_0^{\frac{t}{1-t}} x^2 dx + t\int_0^{\frac{t}{1-t}} x\,dx$

$$= (1-t)\int_0^{\frac{t}{1-t}} x^2\,dx + t\int_0^{\frac{t}{1-t}} x\,dx$$

$$= -(1-t)\left[\frac{x^3}{3}\right]_0^{\frac{t}{1-t}} + t\left[\frac{x^2}{2}\right]_0^{\frac{t}{1-t}}$$

$$= -(1-t)\left(\frac{\frac{t}{1-t}}{3} - 0\right) + t\left(\frac{\left(\frac{t}{1-t}\right)^2}{2} - 0\right)$$

$$= -\frac{1}{6}\,\frac{t^3}{(1-t)^2} + \frac{1}{2}\,\frac{t^3}{(1-t)^2}$$

$$= \frac{t^3}{(1-t)^2}\left(-\frac{1}{6} + \frac{3}{6}\right) = \frac{1}{6}\,\frac{t^3}{(1-t)^2}$$

$$\lambda(t) = \frac{1}{6}\,\frac{t^3}{(1-t)^2}$$

$$\lambda'(t) = 0$$

$$\lambda'(t) = \frac{1}{6}\,\frac{(1-t)^2\cdot 3t^2 - t^3\cdot 2(1-t)}{(1-t)^4}$$

$$= \frac{1}{6}\,\frac{(1-t)3t^2 + 2t^3}{(1-t)^3}$$

$$= \frac{\frac{1}{6}\cdot 3t^2 - 3t^3 + 2t^3}{(1-t)^3}$$

$$= \frac{1}{6}\,\frac{3t^2 - t^3}{(1-t)^3} = 0$$

$$3t^2 - t^3 = 0, \quad 3t^2 = t^3$$

$$t_{1/2} = 0 \qquad t_3 = 3$$

$$\lambda(t=3) = \frac{27}{24}$$